

CS 1671 / CS 2071

Human Language Technologies

Session 5: Machine learning intro, NLP tasks and applications

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September 8, 2026

Course logistics: homework

- [Homework 1](#) has been released
 - **Due next Thu Sep 17 at 11:59pm**
 - Covers text processing and regular expressions in Python
- Homework assignments are programming-based
- Please remind me of your name before asking or answering a question

Course logistics: project

- [Project idea form](#) to submit project ideas you may want to work on
 - Is due **next Tue Sep 15**
- Take a look at the example projects on the [project website](#). You can submit one or more of those for the form, or submit your own idea!
- Have a potential project idea that involves deriving insight from a dataset of text, or building an NLP system that can do something with text. You can submit it!
 - Ideas do not need to be well-formed
 - Ideas that have data already available are more realistic
- You will later choose from an **anonymized** list of project ideas on Project Match Day, Sep 22

Review activity:

Define and discuss
conditional probability

Overview: Machine learning intro, NLP tasks and applications

- Artificial intelligence and machine learning
- Intro to machine learning
 - Definitions
 - Models and algorithms
 - Data: training, development, test
- NLP applications
- NLP “core tasks”
- Coding activity: clickbait classification

Artificial intelligence and machine learning



June 2026,
Penn Ave,
Pittsburgh, PA

Artificial intelligence (AI)

- A broad, umbrella term encompassing technologies that can do tasks considered markers of human “intelligence”
 - Even human “intelligence” is ill-defined
 - Is mostly a marketing term? (Bender and Hanna 2025)
 - Families of technologies including generating and analyzing language (NLP), generating and analyzing images and video (computer vision), playing games and taking actions (reinforcement learning), automated decision-making, recommendation systems
- Machine learning: technique for AI involving training systems to learn patterns from data

• Intro to (supervised) machine learning

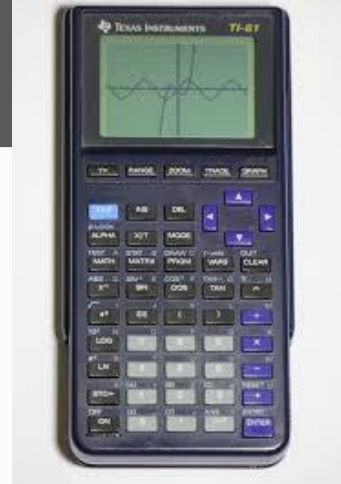
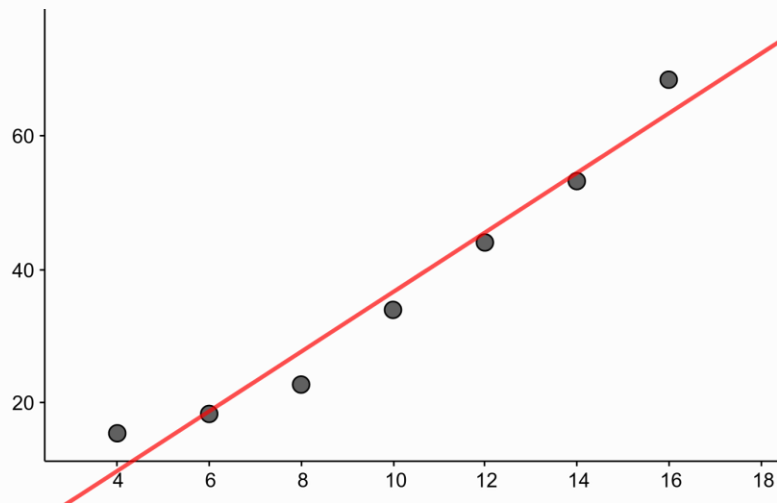
What is machine learning?

- A system that learns a function (maps from an input to an output) from examples/data
- Can predict things and perform tasks **without** explicit instructions
- Learns patterns from data with statistical algorithms
- Examples
 - Predict the weather tomorrow
 - Predict the best next driving action to take in an autonomous vehicle

What can you do with machine learning?

- Prediction: predict an output from an unseen input
 - That fits the pattern learned by looking at input it has seen before
- Interpretation
 - Examine the learned model weights to characterize the relationship between variables

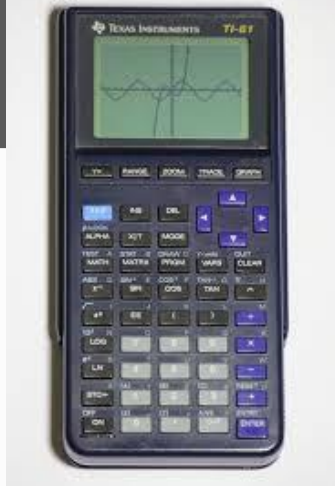
$$y = 4x - 10$$



Machine learning models

- Transform an input to an output with a “model”: a simplified mathematical/statistical version of reality
- Models have parameters **learned from patterns in data**
 - Usually encode how variables relate to each other

$$y = 4x - 10$$



Kimi K3

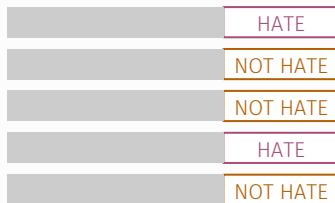
The biggest open model ever. Weights drop July 27.



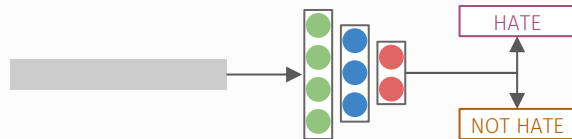
Supervised machine learning process



Data
(input text, X)



Annotate
labels (Y)



Train a model to
predict labels (Y)
from input text (X)

Machine learning algorithms

- Algorithms are systematic ways of doing things
- In machine learning, “algorithms” refers to systematic ways of estimating model parameters from data.
- How does the model learn the patterns that enable it to make predictions? That’s the machine learning algorithm
- We’ll go over many in this class, including:
 - Logistic regression
 - Neural networks
 - Transformers

Training and test sets (and phases)

Training set

Development set

Test set

- Train parameters of the model on training set (training phase)
 - Sees examples of input and (assumed correct) output that it will mimic
- Development set to run tests of the model and choose hyperparameters
- Test time
 - Freeze parameters of the model
 - Predict input from an unseen set
 - Evaluate on correct answers and see how well the model performs
- **Don't look at the test set too much when developing/choosing models**

NLP applications

Core tasks and applications of NLP

APPLICATIONS

machine translation

chatbots

information retrieval

summarization

question answering

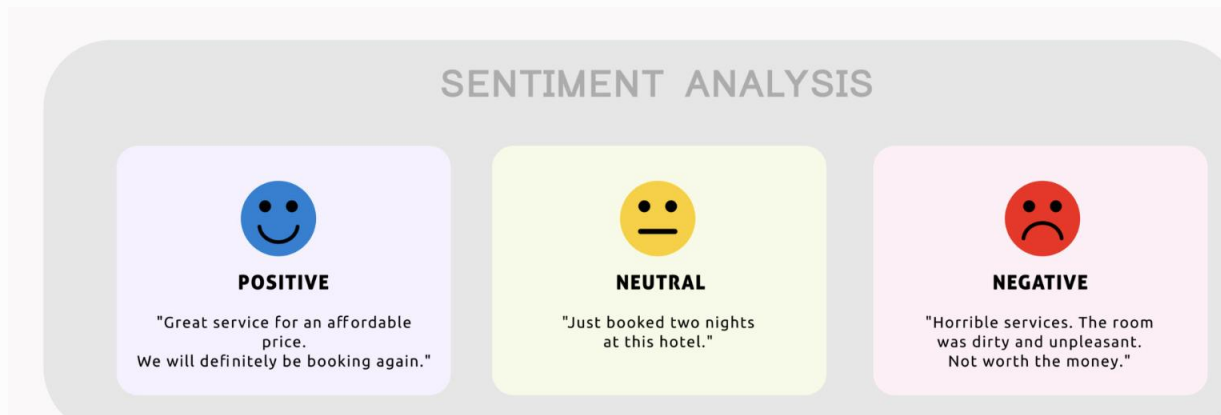
NLP applications: email classification

The screenshot displays a Gmail interface with a sidebar on the left and a main inbox area. The sidebar includes a 'COMPOSE' button, a list of folders (Inbox (7), Starred, Drafts, Sent Mail), and a 'Search people...' section with a list of contacts: Jenny Kang, Peter H, Jonathan Pelleg, Brett C, Max Stein, Jen Hart, and Eric Lowery. The main inbox area is divided into four tabs: Primary, Social (3 new), Promotions (2 new), and Updates (2 new). The Primary tab is selected, showing a list of emails. Each email row includes a checkbox, a star icon, the sender, and the subject. The emails are categorized by sender (Google+, YouTube, or individual names) and their content (tagged photos, video uploads, knitting club, new pup, shared albums, and new apartment). The Social tab is highlighted with a blue bar, and the Promotions and Updates tabs are highlighted with green bars. The Primary tab is highlighted with a grey bar.

Category	Sender	Subject
Primary	Google+	You were tagged in 3 photos on Google+ - Google+ You were tagged in three pl
Primary	YouTube	LauraBlack just uploaded a video. - Jess, have you seen the video LauraBlack u
Primary	Emily Million (Google+)	[Knitting Club] Are we knitting tonight? - [Knitting Club] Are we knitting tonight?
Primary	Sean Smith (Google+)	Photos of the new pup - Sean Smith shared an album with you. View album be thoi
Primary	Google+	Kate Baynham shared a post with you - Follow and share with Kate by adding her
Primary	Google+	Danielle Hoodhood added you on Google+ - Follow and share with Danielle by
Primary	YouTube	Just for You From YouTube: Daily Update - Jun 19, 2013 - Check out the latest
Primary	Google+	You were tagged in 3 photos on Google+ - Google+ You were tagged in three phot
Primary	Hilary Jacobs (Google+)	Check out photos of my new apt - Hilary Jacobs shared an album with you. View
Primary	Google+	Kate Baynham added you on Google+ - Follow and share with Kate by adding her

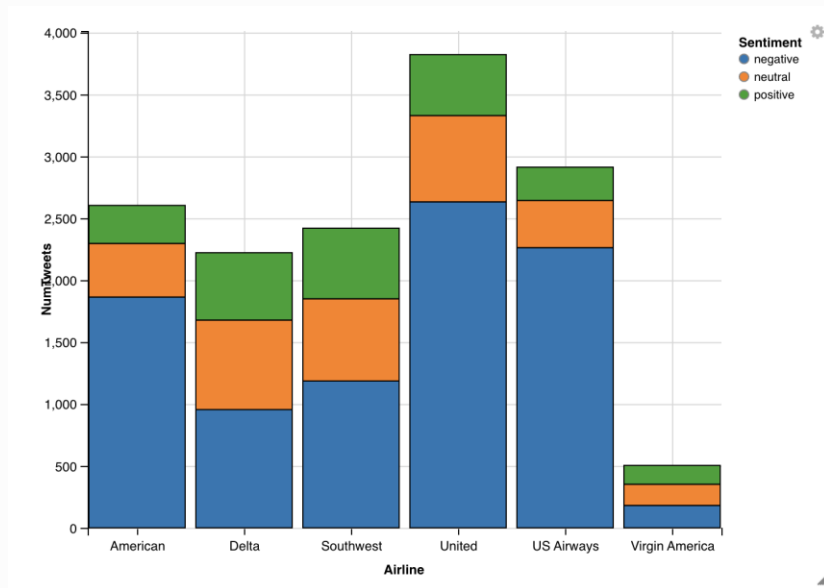
- Spam / Not spam
- Priority Level
- Category (primary / social / promotions / updates)

NLP applications: sentiment analysis



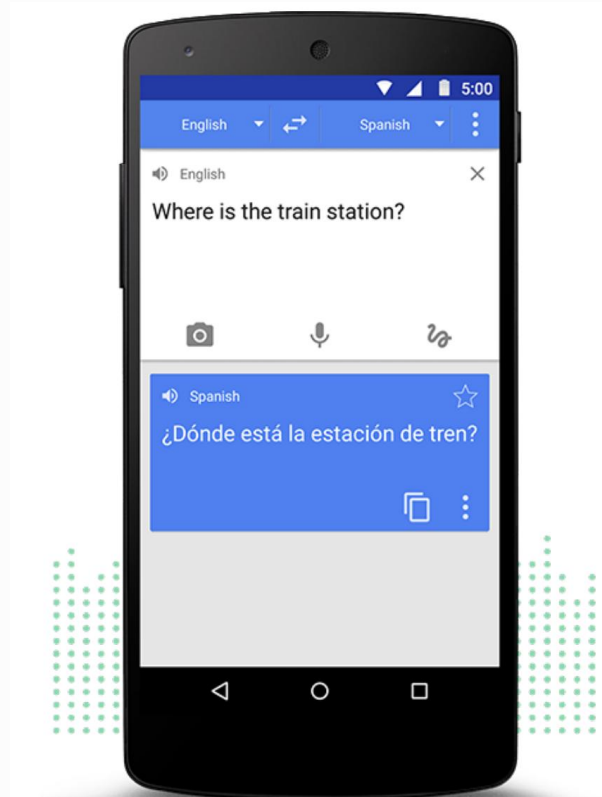
Hotel review sentiment

NLP applications: sentiment analysis



US Airline review sentiment

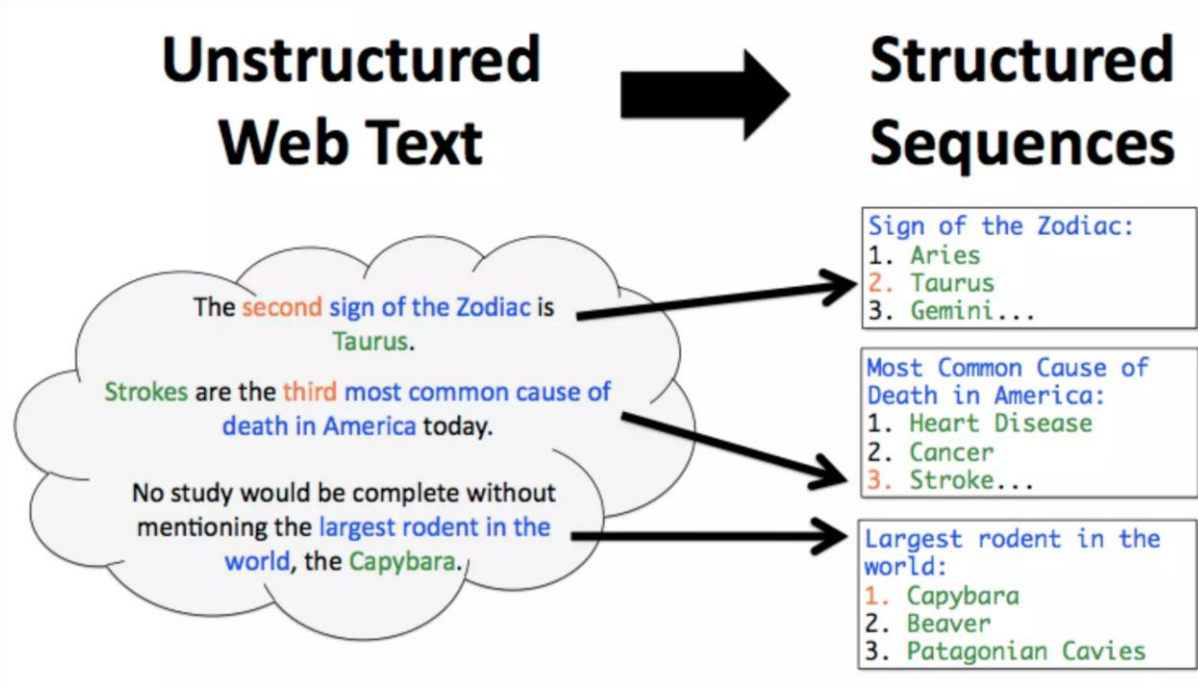
NLP applications: machine translation



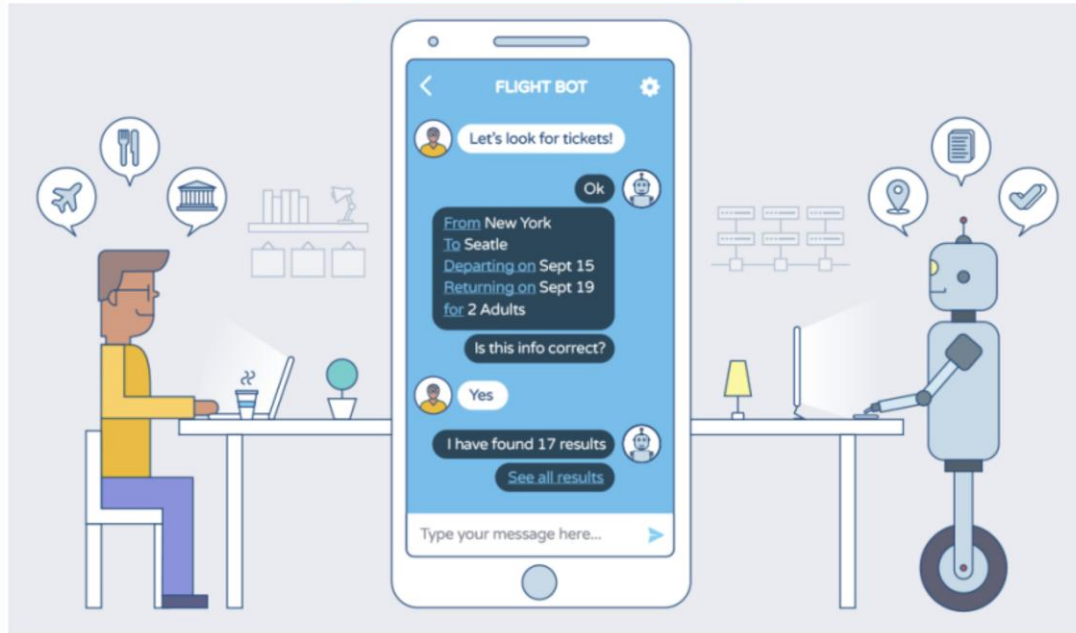
NLP applications: summarization



NLP applications: information extraction



NLP applications: dialogue systems/chatbots



NLP applications: question answering



 amazon alexa

"Alexa, who was President when Barack Obama was nine?"

"Alexa, how's my commute?"

"Alexa, what's the weather?"

"Alexa, did the 49ers win?"



Discuss with a neighbor:

What NLP applications, if any,
do you use?

• NLP core tasks

Core tasks and applications of NLP

CORE TASKS

text classification

language modeling

sequence labeling



APPLICATIONS

machine translation

chatbots

information retrieval

summarization

question answering

Text classification

- Input: a span of text
- Output: a label from a set of discrete options
- *Example:* sentiment analysis
 - *Text* -> {positive, neutral, negative}

Language modeling

- Input: a span of text, or no text at all
- Output: the next word
- *Example:* text generation for chatbots (ChatGPT)
 - *context text -> next word*

Sequence labeling

- Input: a span of text
- Output: a sequence of labels, one for each word (token)
- *Example*: part-of-speech tagging
 - *The book was brilliant -> DET NOUN VERB ADJ*

Coding activity: clickbait classification

Load in-class notebook

1. Go to this [nbgitpuller link](#) (also available on course website)
2. Log in with your Pitt username if necessary
3. Start a server with **TEACH – 6 CPUs, 48 GB**
4. Load custom environment at
`/ix1/cs1671-2026f/class-env/.pixi/envs/default`
5. This should pull the `cs1671_fall2026_jupyterhub` folder into your JupyterLab
6. Open `session5_clickbait_classification.ipynb`